2023-2024

S6 MATH CP

PAPER I

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2023-2024 Secondary Six Final Examination

MATHEMATICS Compulsory Part

PAPER I

Question-Answer Book

24/01/2024

8:15 am – 10:30 am (2¼ hours)

This paper must be answered in English

INSTRUCTIONS

1. After the announcement of the start of the examination, you should first write your Name, Class and Class number in the spaces provided on Page 1.

2. This paper consists of THREE sections, A(1), A(2) and B.

3. Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

4. Graph paper and supplementary answer sheets will be supplied on request. Write your Name, Class and Class number and mark the question number box on each sheet, and fasten them with string INSIDE this book.

5. Unless otherwise specified, all working must be clearly shown.

6. Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures.

7. The diagrams in this paper are not necessarily drawn to scale.

Name: _____

Class: _____( )

	Marker Use Only
Question	Mark
1	
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Total			

SECTION A(1) (35 marks)

1. Simplify $(\alpha^5\beta^{-6})(\alpha\beta^2)^2$ and express your answer in positive indices. (3 marks)

2. Factorize

(a) $6a-9b$,

(b) $8a^2-10ab-3b^2$,

(c) $6a-9b-8a^2+10ab+3b^2$. (4 marks)

Answers written in the margins will not be marked.

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3. Make s the subject of the formula $8(s+3t)=4s+7$. (3 marks)

4. Consider the compound inequality

$$2-\frac{7(x+3)}{8}<-\frac{4x}{5} \quad \text{or} \quad -3-2x\leq 9 \qquad \ldots\ldots\ldots\ldots\ldots(*)$$

(a) Solve (*).

(b) Write down the least integer satisfying (*). (4 marks)

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5. David and Tim have some marbles. The numbers of marbles that David has to that of Tim are in the ratio 2 : 5. If Tim gives 18 marbles to David, they will have the same number of marbles. Find the total number of marbles that David and Tim have. (4 marks)

6. The coordinates of P and Q are (5, 12) and (6, 4) respectively. P is rotated clockwise about the origin through 90° to P'.

(a) Write down the coordinates of P'.

(b) Is $\triangle P'OQ$ a right-angled triangle? Explain your answer. (5 marks)

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7. In Figure 1, $ABCDE$ is a circle. It is given that AC is a diameter of the circle and $CD = DE$.

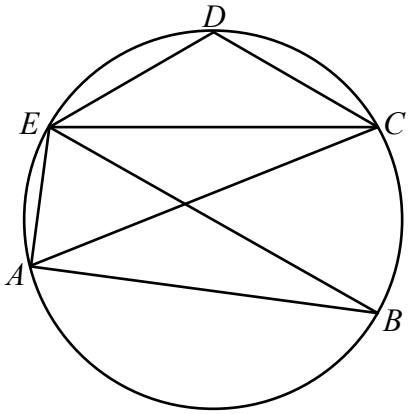


Figure 1

If $\angle ABE = 22°$, find $\angle ACE$ and $\angle DCE$. (4 marks)

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8. Figure 2 shows a trapezium $ABCE$ with $\angle BCE = 90°$ and AE // BC. D is a point on CE such that $\angle ADB = 90°$.

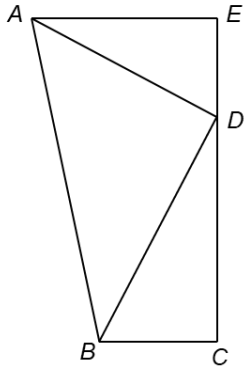


Figure 2

(a) Prove that $\triangle ADE \sim \triangle DBC$.

(b) It is given that $AE = 32$ cm, $CD = 40$ cm and $DE = 18$ cm. Find the area of trapezium $ABCE$.

(4 marks)

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9. The frequency distribution table and the cumulative frequency table below show the javelin results (in m) of a group of 60 students, where a, b, c and d are integers.

Javelin result (m)	Frequency
10 – 14	7
15 – 19	12
20 – 24	a
25 – 29	16
30 – 34	b

Javelin result less than (m)	Cumulative frequency
14.5	7
19.5	c
24.5	30
29.5	d
34.5	60

(a) Find the values of a and b.

(b) Find the mean of the javelin results of the students. (4 marks)

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SECTION A(2) (35 marks)

10. $f(x)$ is partly constant and partly varies as $(x-1)^2$. It is given that $f(0)=6$ and $f(4)=-10$.

(a) Find $f(x)$. (3 marks)

(b) If V is the vertex of the graph of $y=f(x)$, and the graph cuts the x-axis at A and B, find the area of $\triangle VAB$. (4 marks)

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11. The table below shows the distribution of the numbers of times that a group of adults went hiking last year.

Number of times	1	2	3	4	5	6	7
Frequency	6	22	9	2	5	2	4

(a) Find the mean of the distribution. (2 marks)

(b) Are the median and the mode of the distribution equal? Explain your answer. (2 marks)

(c) If x more adults join the group and each of them went hiking 6 times last year, find

(i) the value of x such that the mean of the distribution is increased by 2,

(ii) the least possible value of x such that the median of the distribution is increased by 1,

(iii) the greatest possible value of x such that the mode of the distribution remains unchanged.

(3 marks)

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12. Let $f(x) = 2x^3 - 2x^2 + kx - 126$, where k is a constant. It is given that $f(x) = (x + 3)(ax^2 + bx + c)$, where a, b and c are constants.

(a) Find a, b and c. (4 marks)

(b) Someone claims that all the roots of the equation $f(x) = 0$ are distinct. Do you agree? Explain your answer.

(3 marks)

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13. The base radius of a solid right circular cylinder X and the base radius of the solid right circular cone Y are equal. The heights of X and Y are 20 cm and 12 cm respectively. The volume of the solid right circular cone Z is equal to the sum of the volume of X and the volume of Y. The base radius of Z is equal to the base diameter of X. Peter finds that the volume of Y is 1024π cm^3.

(a) Find the base radius of Y. (1 mark)

(b) Are Y and Z similar? Explain your answer. (3 marks)

(c) Peter claims that the sum of the curved surface area of X and the curved surface area of Y is greater than the curved surface area of Z. Do you agree? Explain your answer. (3 marks)

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14. A is translated 6 units upwards to $B(3, 0)$ and is then reflected about the y-axis to C.

(a) Write down the coordinates of A and C. (2 marks)

(b) Denote the locus of a movable point which maintains a fixed distance of 6 units from B by Γ.

(i) Find the equation of Γ.

(ii) Let D be a point in quadrant III lying on Γ. Find $\angle ADC$. (2 marks)

(c) If P is a moving point such that $AP = PD$ and the equation of the locus of P is $2x - y + k = 0$,

(i) find k,

(ii) find the coordinates of D. (3 marks)

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SECTION B (35 marks)

15. A debate team of 7 students are selected from 5 boys and 8 girls. How many different teams can be formed if

(a) 4 boys are included? (2 marks)

(b) there are more boys than girls in the team? (2 marks)

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16. The equation of the circle C is $x^2 + y^2 + 4x - 96 = 0$. The coordinates of the point A are $(4, -8)$.

(a) Show that A lies on C. (1 mark)

(b) It is given that the straight lines L_1: $3x - 4y - 44 = 0$ and L_2 are tangents to C at the points A and B respectively. L_2 is perpendicular to L_1, and cuts L_1 and the positive y-axis at the points D and E respectively. P is a moving point in the rectangular coordinate plane such that the perpendicular distance from P to DA is equal to the perpendicular distance from P to DB. Denote the locus of P by Γ.

(i) Describe the geometric relationship between Γ and $\angle ADB$.

(ii) Find the equation of Γ. (6 marks)

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17. Let k be a positive constant. The roots of the equation $20x^2 - 4kx + k = 0$ are α and β.

(a) Express $\alpha + \beta$ and $\alpha\beta$ in terms of k. (1 mark)

(b) Let $A(n)$ be the nth term of a geometric sequence. It is given that

$$A(1) = \alpha\beta,\ A(2) = 0.9 \text{ and } A(3) = \alpha + \beta.$$

Suppose that $\log_3 B(n) = A(n)$ for any positive integer n. Find the least value of m such that $\log_9 (B(1)\ B(2)\ B(3)\ \ldots\ B(m)) > 202\ 400\ 000$. (5 marks)

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18. (a) It is given that ABC is a triangle, where $AB = 10$ cm, $BC = 8$ cm and $AC = 12$ cm. Find $\angle ABC$.

(2 marks)

(b) Figure 3 shows the pyramid $ABCY$. The base of this pyramid is the triangle ABC described in (a). P, Q and R are points lying on AB, BC and CA respectively such that YP, YQ and YR are altitudes of the triangles YAB, YBC and YAC respectively. It is given that $YP = YQ = YR$.

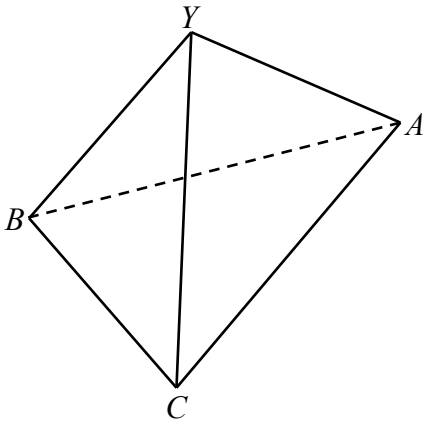


Figure 3

It is given that the angle between the triangles YAC and ABC is 55°. Someone claims that the angle between YB and the triangle ABC exceeds 40°. Is the claim correct? Explain your answer.

(4 marks)

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19. The coordinates of the points A and B are $(0 , 4)$ and $(32 , -20)$ respectively. C is a point in the rectangular coordinate plane such that BC is a vertical line. H is a point in the same rectangular coordinate plane such that $\angle AHB = \angle AHC$ and $BH = CH$.

(a) Prove that $\triangle ABH \cong \triangle ACH$. (2 marks)

(b) Find the coordinates of C. (2 marks)

(c) Suppose that $AH = BH = CH$. Denote the in-centre of $\triangle ABC$ by I.

(i) Find the coordinates of the circumcentre of $\triangle ABC$.

(ii) Are A, H and I collinear? Explain your answer.

(iii) Someone claims that the area of the circumcircle of $\triangle ABC$ is greater than 4 times the area of the inscribed circle of $\triangle ABC$. Do you agree? Explain your answer.

(8 marks)

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